

ZILLOW DATASET: ASSESSED TAX VALUE MODELING

OMCDS MOD 3

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Executive Summary

This project is aimed at producing accurate predictions for tax-assessed value of homes based on the Zillow Housing Dataset.

- Includes features like square footage, location, bed/bathrooms, and more
- Performed cleaning, preprocessing, and feature engineering on the data
- Decided on Gradient Boosting Regressor after testing a number of different models

Result:

A reliable and interpretable model that can support homeowners, real estate investors, and tax authorities with property valuations.

In today's data-driven real estate market, accurate property valuation models are essential for improving transparency and fairness in tax assessments. By developing a model that enhances Zillow's existing valuation pipeline, we aim to benefit a broad range of stakeholders—including homeowners, investors, tax authorities, and real estate professionals—by improving the accuracy, interpretability, and fairness of automated property tax assessments.

How do we accomplish this?

- Robust data cleaning and reduction
- Development of new features

How do we measure success?

- RMSE, MAE, and R^2 on unseen data
- Generalization power and interpretability

Introduction



The Data

The 2016 Zillow Property Dataset:

- 3 million property records
- Various U.S. regions
- home characteristics, geographic identifiers, and administrative data.

Key characteristics:

- Size: ~77,000 rows before filtering; reduced to ~63,000,000 after cleaning
- Target Variable: taxvaluedollarcnt — the Zillow-assessed property tax value
- Feature Types: Numerical categorical and string-based
- Missing Values: Some >80%, so we applied strict thresholds to retain only well-populated features.
- Redundancy: Location-based features were streamlined to avoid duplication.

	parcelid	airconditioningtypeid	architecturalstyletypeid	basementsqft	bathroomcnt	bedroomcnt	buildingclasstypeid	buildingqualitytypeid	calculatedbathnbr	decktypeid	...
0	14297519	NaN	NaN	NaN	3.5	4.0	NaN	NaN	3.5	NaN	...
1	17052889	NaN	NaN	NaN	1.0	2.0	NaN	NaN	1.0	NaN	...
2	14186244	NaN	NaN	NaN	2.0	3.0	NaN	NaN	2.0	NaN	...
3	12177905	NaN	NaN	NaN	3.0	4.0	NaN	8.0	3.0	NaN	...
4	10887214	1.0	NaN	NaN	3.0	3.0	NaN	8.0	3.0	NaN	...

RangeIndex: 77613 entries, 0 to 77612			
Data columns (total 55 columns):			
#	Column	Non-Null Count	Dtype
0	parcelid	77613 non-null	int64
1	airconditioningtypeid	25007 non-null	float64
2	architecturalstyletypeid	207 non-null	float64
3	basementsqft	50 non-null	float64
4	bathroomcnt	77579 non-null	float64
5	bedroomcnt	77579 non-null	float64
6	buildingclasstypeid	15 non-null	float64
7	buildingqualitytypeid	49809 non-null	float64
8	calculatedbathnbr	76963 non-null	float64
9	decktypeid	614 non-null	float64
10	finishedfloorisquarefeet	6037 non-null	float64
11	calculatedfinishedsquarefeet	77378 non-null	float64
12	finishedsquarefeet12	73923 non-null	float64
13	finishedsquarefeet13	42 non-null	float64
14	finishedsquarefeet15	3027 non-null	float64
15	finishedsquarefeet50	6037 non-null	float64
16	finishedsquarefeet6	386 non-null	float64
17	fips	77579 non-null	float64
18	fireplacecnt	8289 non-null	float64
19	fullbathcnt	76963 non-null	float64
20	garagecarcnt	25520 non-null	float64
21	garagetotalsqft	25520 non-null	float64
22	hashottuborspa	1539 non-null	object
23	heatingorsystemtypeid	49571 non-null	float64
24	latitude	77579 non-null	float64
25	longitude	77579 non-null	float64
26	lotssizesquarefeet	69321 non-null	float64
27	poolcnt	16174 non-null	float64
28	poolsum	869 non-null	float64
29	pooltypeid10	465 non-null	float64
...			
53	censustractandblock	77332 non-null	float64
54	taxvaluedollarcnt	77578 non-null	float64

After Cleaning and Engineering

RangeIndex: 62814 entries, 0 to 62813			
Data columns (total 25 columns):			
#	Column	Non-Null Count	Dtype
0	bathroomcnt	62814 non-null	int64
1	bedroomcnt	62814 non-null	int64
2	buildingqualitytypeid	62814 non-null	float64
3	calculatedfinishedsquarefeet	62814 non-null	float64
4	finishedsquarefeet12	62814 non-null	float64
5	garagecarcnt	62814 non-null	int64
6	garagetotalsqft	62814 non-null	float64
7	lotssizesquarefeet	62814 non-null	float64
8	poolcnt	62814 non-null	float64
9	roomcnt	62814 non-null	int64
10	unitcnt	62814 non-null	int64
11	yearbuilt	62814 non-null	float64
12	numberofstories	62814 non-null	float64
13	fips_6037.0	62814 non-null	float64
14	fips_6059.0	62814 non-null	float64
15	fips_6111.0	62814 non-null	float64
16	regionidcounty_1286.0	62814 non-null	float64
17	regionidcounty_2061.0	62814 non-null	float64
18	regionidcounty_3101.0	62814 non-null	float64
19	log_garagetotalsqft	62814 non-null	float64
20	log_lotssizesquarefeet	62814 non-null	float64
21	log_calculatedfinishedsquarefeet	62814 non-null	float64
22	calc_bathbath	62814 non-null	float64
23	prod_bathroomcnt_bedroomcnt	62814 non-null	int64
24	taxvaluedollarcnt	62814 non-null	float64

METHODOLOGY



Analytical Framework

RMSE (Root Mean Squared Error) was chosen for its sensitivity to large errors, aligning with the business goal of accurate high-value home assessments.

← Error Metric

We used cross-validation to assess generalization, particularly KFold and RepeatedKFold, to ensure stability across multiple data splits.

← Validation

We explored linear models, tree-based models, and ensemble techniques to cover a range of complexities and interpretabilities.

← Model Selection

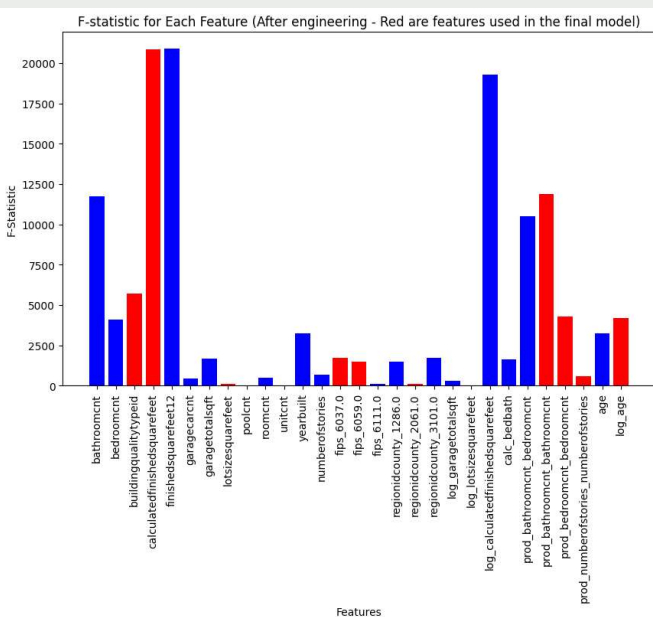
Initially, RandomizedSearchCV was used for efficient exploration. GridSearchCV was later removed due to marginal performance gains and high computational cost.

← Hyperparameter Tuning

Step	Method
1: Initial Cleanup	Dropping columns with no predictive power or high redundancy
2: Handling Missing Data	Dropped columns with 80% null and rows with 25% null
3: Remove Outliers	Reduced the dataset by 20% by trimming according to the IQR
4: Imputation	Median fill for numeric columns and most common value for categorical
5: Final	One-Hot Encoding, Rounding count columns, and X/y split

Cleaning and Preprocessing

5 Steps



Feature Engineering

Bed-to-Bath Ratio: $\text{bedroomcnt} / (\text{bathroomcnt} + 1)$

Bed-to-Bath Product: bathroomcnt * bedroomcnt

Age Feature: Formula: 2025 - yearbuilt

Polynomial Features: Squares of various features to increase their variance on the higher end

Log Transformations: Logarithms to eliminate right-skewed features' effects

Model Evaluation Approach

- Plot validation curves to monitor overfitting or underfitting under various hyperparameter configurations
- Track performance across multiple metrics (RMSE, R^2 , training time).

Category	Model(s)	Best
Linear	Linear Regression, Ridge, Lasso	Ridge
Tree	DecisionTreeRegressor, RandomForestRegressor	RF
Ensemble	GradientBoostingRegressor	GBR

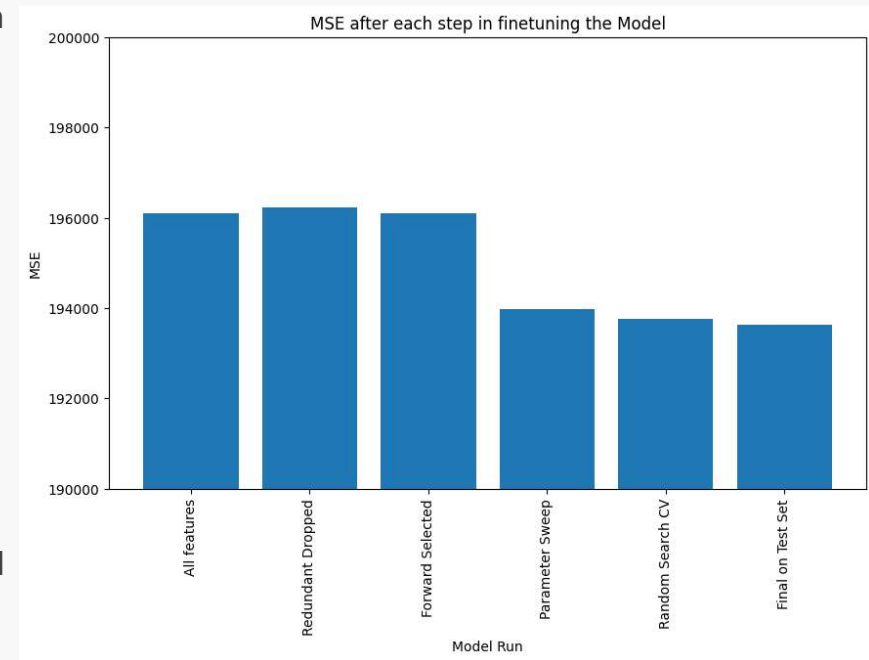
Model Selection: GradientBoostingRegressor

Results and Evaluation

We selected the GradientBoostingRegressor (GBR) as our final model based on its strong overall performance, generalization capability, and ability to model non-linear relationships.

- Train RMSE: 189,200.81
- Test RMSE: 193,634.37
- Cross-Validated RMSE: 193,744.46
- CV RMSE Std Dev: 502,332.56
- Test R^2 Score ~ 0.43 – 0.48 (est.)

Although Ridge Regression remained a strong contender (offering faster training and reasonably good test performance), it consistently underperformed the GBR in final runs by a measurable margin.



ERROR ANALYSIS

- **Outliers:** Many high-value homes were harder to predict — likely due to missing luxury indicators not captured in the dataset.
- **Feature Sensitivity:** GBR leveraged engineered features like Bed-to-Bath ratio and Age more effectively than simpler models.
- **Overfitting Risk:** Controlled by early stopping and regularization in GBR, with validation curves confirming minimal overfit.

Conclusions

Findings:

- The importance of rigorous cleaning and engineering
- Ensemble models advantages with nonlinear data
- Our model can support fair and quick tax assessments for a range of stakeholders

Limitations:

- The lack of certain key features (e.g., renovation, neighborhood amenities)
- High value homes remain difficult to value

Future:

- Incorporate more datasets
- Experiment with XGBoost for faster training
- Develop a lightweight version for web integration

The background is a minimalist geometric design. It features a grid of squares in various shades of gray, blue, and purple. A central dark purple rectangle contains the text "THANK YOU" in white, bold, sans-serif capital letters. The overall aesthetic is modern and clean.

THANK YOU